Ideal and Non-ideal solution

1. Which of the following liquid pairs shows a positive deviation from Raoult's law
(a) Water-nitric acid
(b) Benzene-methanol
(c) Water-hydrochloric acid
(d) Acetone-chloroform

2. Which one of the following is non-ideal solution
(a) Benzene + toluene
(b) n -hexane + n -heptane
(c) Ethyl bromide + ethyl iodide
(d) $CCl_4 + CHCl_3$

3. A non ideal solution was prepared by mixing 30 ml chloroform and 50 ml acetone. The volume of mixture will be
(a) > 80 ml (b) < 80 ml
(c) = 80 ml (d) ≥ 80 ml

4. Which pair from the following will not form an ideal solution
(a) $CCl_4 + SiCl_4$
(b) $H_2O + C_4H_9OH$
(c) $C_2H_5Br + C_2H_5I$
(d) $C_6H_{14} + C_7H_{16}$

5. An ideal solution is that which
(a) Shows positive deviation from Raoult's law
(b) Shows negative deviation from Raoult's law
(c) Has no connection with Raoult's law
(d) Obeys Raoult's law

6. Which one of the following mixtures can be separated into pure components by fractional distillation
(a) Benzene – toluene
(b) Water – ethyl alcohol
(c) Water – nitric acid
(d) Water – hydrochloric acid

7. All form ideal solutions except
(a) C_2H_5Br and C_2H_5I
(b) C_6H_5Cl and C_6H_5Br
(c) C_6H_6 and $C_6H_5CH_3$
(d) C_2H_5I and C_2H_5OH

8. Which property is shown by an ideal solution
(a) It follows Raoult's law
(b) $\Delta H_{mix} = 0$
(c) $\Delta V_{mix} = 0$
(d) All of these

9. When two liquid A and B are mixed then their boiling points becomes

greater than both of them. What is the nature of this solution

(a) Ideal solution
(b) Positive deviation with non ideal solution
(c) Negative deviation with non ideal solution
(d) Normal solution

10. In mixture *A* and *B* components show –*ve* deviation as

(a) $\Delta V_{\text{mix}} > 0$
(b) $\Delta H_{\text{mix}} < 0$
(c) *A-B* interaction is weaker than *A-A* and *B-B* interaction
(d) *A-B* interaction is strong than *A-A* and *B-B* interaction

11. In which case Raoult's law is not applicable

(a) $1MNaCl$
(b) 1 M urea
(c) 1 M glucose
(d) 1 M sucrose

12. A solution that obeys Raoult's law is

(a) Normal (b) Molar
(c) Ideal (d) Saturated

13. An example of near ideal solution is

(a) n -heptane and n -hexane
(b) $CH_3COOH + C_5H_5N$
(c) $CHCl_3 + (C_2H_5)_2O$
(d) $H_2O + HNO_3$

14. A mixture of liquid showing positive deviation in Raoult's law is

(a) $(CH_3)_2CO + C_2H_5OH$
(b) $(CH_3)_2CO + CHCl_3$
(c) $(C_2H_5)_2O + CHCl_3$
(d) $(CH_3)_2CO + C_6H_5NH_2$

15. All form ideal solution except

(a) C_2H_5Br and C_2H_5I
(b) C_2H_5Cl and C_6H_5Br
(c) C_6H_6 and $C_6H_5CH_3$
(d) C_2H_5I and C_2H_5OH

16. Formation of a solution from two components can be considered as

(i) Pure solvent $\rightarrow$ separated solvent molecules ΔH_1
(ii) Pure solute $\rightarrow$ separated solute molecules ΔH_2
(iii) Separated solvent and solute molecules $\rightarrow$ solution ΔH_3

Solution so formed will be ideal if

(a) $\Delta H_{\text{soln}} = \Delta H_3 - \Delta H_1 - \Delta H_2$
(b) $\Delta H_{\text{soln}} = \Delta H_1 + \Delta H_2 + \Delta H_3$
(c) $\Delta H_{\text{soln}} = \Delta H_1 + \Delta H_2 - \Delta H_3$
(d) $\Delta H_{\text{soln}} = \Delta H_1 - \Delta H_2 - \Delta H_3$

17. Identify the mixture that shows positive deviation from Raoult's law

(a) $CHCl_3 + (CH_3)_2CO$
(b) $(CH_3)_2CO + C_6H_5NH_2$

(c) $CHCl_3 + C_6H_6$

(d) $(CH_3)_2CO + CS_2$

(e) $C_6H_5N + CH_3COOH$

18. When acetone is added to chloroform, then hydrogen bond is formed between them.These liquids show

(a) Positive deviation from Raoult's law

(b) Negative deviation from Raoult's law

(c) No deviation from Raoult's law

(d) Volume is slightly increased

19. Which of the following is true when components forming an ideal solution are mixed

(a) $\Delta H_m = \Delta V_m = 0$

(b) $\Delta H_m > \Delta V_m$

(c) $\Delta H_m < \Delta V_m$

(d) $\Delta H_m = \Delta V_m = 1$

20. The liquid pair benzene-toluene shows

(a) Irregular deviation from Raoult's law

(b) Negative deviation from Raoult's law

(c) Positive deviation from Raoult's law

(d) Practically no deviation from Raoult's law